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RELEASE LEVER MECHANISM FOR SERVER CHASSIS SLEDS

Abstract

A compact lever mechanism for assisting in removing and inserting a sled in a chassis is disclosed. The lever mechanism has a longitudinal lever member. A spur gear having gear teeth is coupled to one end of the longitudinal lever member. An ejector component is rotatably connected to the sled. The ejector component has a frame defined by an internal gear support, a first support arm, and a second support arm. The spur gear is positioned within the frame and engages the internal gear support to rotate the ejector component. The first support arm includes a tab engaging a flange of the chassis to apply force to remove the sled from the chassis when rotated in a first direction. The second support arm includes a second tab engaging a cutout in the chassis to apply force to insert the sled in the chassis when rotated in a second direction.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to lever release mechanisms for sleds of computing devices. More particularly, aspects of this disclosure relate to a space saving lever mechanism that facilitates insertion and removal of device sleds from a chassis.

BACKGROUND

[0002] Computer systems (e.g., desktop computers, blade servers, rack-mount servers, etc.) are employed in large numbers in various applications. High demand applications, such as network based systems, data centers, or high density finite element simulations can push hardware of computing systems, and require servers with specialized capabilities. Modern server chassis designs allow flexibility in terms of capabilities and components. One method is to design bays that allow server sleds to be easily installed and be coupled to common network and power interfaces. Such sleds include processors, memories, and support circuits for a server on a tray or bottom panel. The back end of the sled typically includes connectors that are mated with compatible connectors on the chassis for power and electronic signals. In order for easy service of and/or changing components, sleds may be accessed without having to move the actual server chassis to access the components on the sled. For example, certain sleds may slide out from the chassis, allowing the installation or removal of a component on the sled or the replacement of the sled entirely. Once installation is complete, the sled may be pushed back into the chassis and resume operation. Thus, a sled will typically include release and locking mechanisms on a front end that allows the sled to be inserted and removed from the bay.

[0003] Currently sleds use mechanical devices to assist in providing force to ensure that the sleds are properly inserted so the connectors are plugged into the chassis. Conversely, the mechanical devices are used to provide additional force to unplug the connectors when the sleds are pulled out. For example, known sleds have rotatable levers that allow an operator to rotate the lever to an open position to provide force to disconnect the sled and allow the sled to be pulled out from the chassis. When the sled is pushed into the chassis, the lever is rotated to the closed position to provide force to plug in the connectors and fully insert the sled into the chassis. Thus, existing lever designs are a mechanical system where the lever rotates around a pin to translate force to disconnect various connectors between the sled to the chassis.

[0004] FIG. 1A and FIG. 1B show perspective views of a server chassis **10** with a sled **12**. The chassis **10** has side walls **20** and **22** that define a bay for the sled **12**. The end of the bay includes various connectors for connecting the sled **12** to power sources and data components of the chassis **10**. The side walls **20** and **22** each have an edge element **24**. The sled **12** includes known levers **30** and **32** that are used to assist in releasing the sled **12** from connection to a chassis **10**. Rotating the levers such as the lever **30** shown in FIG. 1B provide force allowing the sled **12** to be pushed away from the chassis **10** through contact of an engagement end **40** of the lever **30** with the edge element **24** of the side wall **20**. An opposite end **42** allows a user to grip the lever **30** to rotate the lever **30** and thus move the sled **12** out of the bay in the chassis **10**.

[0005] FIG. 1C is a perspective view of the known lever **30**. The lever **30** includes a main body **50** that defines the engagement end **40** and the opposite end **42**. A handle **52** is attached to the opposite end **42** to allow a user to better engage the lever **30** to move the lever. The engagement end **40** includes a hole **54** that allows the insertion of a pin to create a hinge around which the lever **30** may be rotated. A tab **56** contacts the edge **24** to apply the force of rotating the lever **30** to linear force to push the sled **12** away from the chassis **10**. An opposite tab **58** translates the force from the rotation of the lever **30** in the opposite direction to linear force to push the sled **12** into the chassis **10**.

[0006] The known lever **30** provides force relative to the side wall **20** proportional to the length of the main body **50**. Thus, such levers need to be longer to provide more force to insert or retract a sled. However, longer levers require more space on the front of the sled.

[0007] Thus, there is a need for a compact lever release mechanism that exerts sufficient force to

attach or release a sled. There is a further need for a lever release mechanism that uses multiple gears to more efficiently use force applied by the lever.

SUMMARY

[0008] One disclosed example is a lever mechanism for insertion to and removal of a sled from a chassis. The lever mechanism has a longitudinal lever member having a release end and an opposite pivot end. A spur gear having gear teeth is coupled to the pivot end of the lever. The spur gear is rotated by rotating the longitudinal lever member. An ejector component is rotatably connectable to the sled. The ejector component has a frame defined by an internal gear support, a first support arm, and a second support arm. The spur gear is positioned within the frame and engages the internal gear support to rotate the ejector component. The first support arm includes a tab engaging a flange of the chassis to apply force to remove the sled from the chassis when the longitudinal lever member is rotated in a first direction. The second support arm includes a second tab engaging a cutout of the chassis to apply force to insert the sled in the chassis when the longitudinal lever member is rotated in a second direction.

[0009] A further implementation of the example lever mechanism is an embodiment including a support plate rotatably attached to the spur gear and ejector component. A spring has one end connected to the support plate and an opposite end connected to the spur gear to bias the spur gear in the first rotational direction. Another implementation is where the release end of the longitudinal lever member includes a tab that engages a latch mechanism on the sled to hold the longitudinal lever member in place. Another implementation is where the latch mechanism rotates between a position engaging the tab and a position releasing the tab. Another implementation is where the release end of the longitudinal lever member includes a handle. Another implementation is where the spur gear is connected to a cylindrical support, and the longitudinal lever member is connected to the cylindrical support. Another implementation is where the longitudinal lever member, spur gear, and ejection component generate force to disconnect connectors of the sled from connectors of the chassis when rotated in the first direction.

[0010] Another disclosed example is a sled for insertion into a bay of a chassis defined by two edge members each having a flange and a cutout. The sled includes a tray holding electronic components. A front panel is coupled perpendicularly to the tray. A first lever mechanism is mounted on a side of the front panel. The first lever mechanism includes a longitudinal lever member having a release end and an opposite pivot end. A spur gear having gear teeth is coupled to the pivot end of the longitudinal lever member. The spur gear is rotated by rotating the longitudinal lever member. An ejector component is rotatably connected to the sled. The ejector component has a frame defined by an internal gear support, a first support arm, and a second support arm. The spur gear is positioned within the frame and engages the internal gear support to rotate the ejector component. The first support arm includes a tab engaging one of the flanges to apply force to remove the sled from the bay when the longitudinal lever member is rotated in a first direction. The second support arm includes a second tab engaging one of the cutouts to apply force to insert the sled in the bay when the longitudinal lever member is rotated in a second direction.

[0011] A further implementation of the example sled is an embodiment including a second lever mechanism mounted on an opposite side of the front panel. The second lever mechanism includes a longitudinal lever member having a release end and an opposite pivot end. A spur gear having gear teeth is coupled to the pivot end of the longitudinal lever member. An ejector component is rotatably connected to the sled. The ejector component has a frame defined by an internal gear support, a first support arm, and a second support arm. The spur gear is positioned within the frame and engages the internal gear support to rotate the ejector component. The first support arm includes a tab engaging the other one of the flanges to apply force to remove the sled from the bay when the longitudinal lever member is rotated in a first direction. The second support arm includes a second tab engaging the other one of the cutouts to apply force to insert the sled in the bay when the longitudinal lever member is rotated in a second direction. Another implementation is where the

first and second lever mechanisms generate sufficient force to disconnect connectors of the sled from connectors of the bay when rotated in the first direction. Another implementation is where the sled includes a support plate rotatably attached to the spur gear and ejector component. A spring has one end connected to the support plate and an opposite end connected to the spur gear to bias the spur gear in the first rotational direction. Another implementation is where the sled further includes a latch mechanism on the front panel. The release end of the longitudinal lever member includes a tab that engages the latch mechanism to hold the longitudinal lever member in place. Another implementation is where the latch mechanism rotates between a position engaging the tab and a position releasing the tab. Another implementation is where the release end of the longitudinal lever member includes a handle. Another implementation is where the spur gear is connected to a cylindrical support, and wherein the longitudinal lever member is connected to the cylindrical support.

[0012] Another disclosed example is a computer system with comprising a chassis having two side walls and a shelf between the two side walls. A bay is defined by the two side walls and the shelf. Each of the side walls have edge members with longitudinal flanges. A sled is insertable in the bay of the chassis between the two side walls. The sled includes a front panel, a tray, and a first rotating lever mechanism on the front panel. The first rotating lever mechanism includes a longitudinal lever member having a release end and an opposite pivot end. A spur gear has gear teeth coupled to the pivot end of the longitudinal lever member. An ejector component is rotatably connected to the sled. The ejector component has a frame defined by an internal gear support, a first support arm, and a second support arm. The spur gear is positioned within the frame and engages the internal gear support to rotate the ejector component. The first support arm includes a tab engaging one of the longitudinal flanges to apply force to remove the sled from the bay when the longitudinal lever member is rotated in a first direction. The second support arm includes a second tab engaging a cutout on one of side walls to apply force to insert the sled in the bay when the longitudinal lever member is rotated in a second direction.

[0013] The above summary is not intended to represent each embodiment or every aspect of the present disclosure. Rather, the foregoing summary merely provides an example of some of the novel aspects and features set forth herein. The above features and advantages, and other features and advantages of the present disclosure, will be readily apparent from the following detailed description of representative embodiments and modes for carrying out the present invention, when taken in connection with the accompanying drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The disclosure will be better understood from the following description of embodiments together with reference to the accompanying drawings.

[0015] FIG. 1A is a front view of a server component with a prior art release lever;

[0016] FIG. 1B is a close up perspective view of the prior art release lever in an open position;

[0017] FIG. 1C is an isolated perspective view of the prior art release lever in FIG. 1A;

[0018] FIG. 2A is a perspective view of a computer system with example release lever mechanisms on a sled, according to certain aspects of the present disclosure;

[0019] FIG. 2B is a close up perspective view of one of the example lever mechanisms in FIG. 2A in a closed position, according to certain aspects of the present disclosure;

[0020] FIG. 2C is a perspective view of the computer system in FIG. 2A with the example release lever mechanisms in an open position, according to certain aspects of the present disclosure;

[0021] FIG. 2D is a close up perspective view of one of the example lever mechanisms in FIG. 2A in an open position, according to certain aspects of the present disclosure;

[0022] FIG. 3 is a bottom view of one of the example lever mechanisms in FIG. 2A, according to certain aspects of the present disclosure;

[0023] FIG. 4 is a close up view of the internal spur and internal gears of the example lever mechanism in FIG. 3, according to certain aspects of the present disclosure;

[0024] FIG. 5 is perspective view of one of the example lever mechanisms in FIG. 2A, according to certain aspects of the present disclosure;

[0025] FIG. 6A-6F is series of views showing the positioning of the example lever mechanism in FIG. 2A between the open and closed positions, according to certain aspects of the present disclosure;

[0026] FIG. 7A is a perspective view of the latch mechanism of the example lever mechanism in FIG. 2A, according to certain aspects of the present disclosure;

[0027] FIG. 7B is an exploded perspective view of the latch mechanism of the example lever mechanism in FIG. 2A, according to certain aspects of the present disclosure;

[0028] FIGS. 8A-8B are views of the latch mechanism in FIG. 7A in locked and unlocked positions, according to certain aspects of the present disclosure;

[0029] FIGS. 9A-9C show the sequence of removing a sled from the server chassis using the example lever mechanisms, according to certain aspects of the present disclosure; and

[0030] FIGS. 10A-10C show the sequence of inserting a sled into the server chassis using the example lever mechanisms, according to certain aspects of the present disclosure.

[0031] The present disclosure is susceptible to various modifications and alternative forms. Some representative embodiments have been shown by way of example in the drawings and will be described in detail herein. It should be understood, however, that the invention is not intended to be limited to the particular forms disclosed. Rather, the disclosure is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

[0032] The present inventions can be embodied in many different forms. Representative embodiments are shown in the drawings, and will herein be described in detail. The present disclosure is an example or illustration of the principles of the present disclosure, and is not intended to limit the broad aspects of the disclosure to the embodiments illustrated. To that extent, elements and limitations that are disclosed, for example, in the Abstract, Summary, and Detailed Description sections, but not explicitly set forth in the claims, should not be incorporated into the claims, singly or collectively, by implication, inference, or otherwise. For purposes of the present detailed description, unless specifically disclaimed, the singular includes the plural and vice versa; and the word “including” means “including without limitation.” Moreover, words of approximation, such as “about,” “almost,” “substantially,” “approximately,” and the like, can be used herein to mean “at,” “near,” or “nearly at,” or “within 3-5% of,” or “within acceptable manufacturing tolerances,” or any logical combination thereof, for example.

[0033] The present disclosure relates to a lever release mechanism for a component sled that uses an internal spur gear in conjunction with gearing of an ejector structure to assist in inserting or releasing the sled from a bay of a chassis. The ejector structure has a frame that contains the internal spur gear. Both the ejector structure and internal spur gear are rotationally mounted on a mounting plate of the sled. The example lever release mechanism generates greater force from leveraging the spur gear driving an internal gear of the ejector structure. Due to the use of multiple gears to efficiently translate rotational force, the example lever release mechanism is more compact than known release levers.

[0034] FIG. 2A shows a perspective view of a computer system **100** that includes a server chassis **110** with a sled **200**. The sled **200** has a roughly rectangular structure that includes a roughly rectangular tray or bottom panel. The sled holds a motherboard that supports one or more components such as power supplies, processors such as CPUs, network interface cards, memory

devices, and the like. The chassis **110** has side walls **120** and **122**. The side walls are joined by a top panel **124** and a shelf **126** that define a bay **128** for the sled **200**. The end of the bay **128** includes various connectors (not shown) for connecting the sled **200** to power sources and data components of the chassis **110**. The side wall **122** has an edge member **130**. The edge member **130** includes a lateral flange **132** that extends the length of the chassis **110**. A corresponding edge member (not shown) is provided for the side wall **120**. The edge member **130** provides additional structural strength for the chassis **110**.

[0035] FIG. 2B is a close up perspective view of the sled **200** when fully inserted into the bay **128** of the chassis **110**. The sled **200** includes a front end **210** that has a series of panels including an upper panel **212**. Each of the panels of the front end **210** such as the upper panel **212** have ventilation holes to allow air flow to the internal components on the motherboard of the sled **200**. An opposite rear end (not shown) of the sled **200** has a series of connectors that, when connected to corresponding connectors at the rear of the bay **128**, receive power and data signals from the chassis **110**.

[0036] As shown in FIG. 2A, the front end **210** of the sled **200** includes lever mechanisms **230** and **232** that are used to assist in releasing the sled **200** from the bay **128** of the chassis **110**. A cutout **134** is positioned on the edge members **130** in alignment with the example lever mechanisms **230** and **232**. The lever mechanisms **230** and **232** are shown in a closed position in FIG. 2A when the sled **200** is entirely inserted in the bay **128**. The lever mechanisms **230** and **232** are identical and thus their respective components will be labeled with common reference numerals in the description herein. The lever mechanisms **230** and **232** may be rotated outward to provide force to allow the sled **200** to be pulled away from the chassis **110** as shown in FIG. 2C. FIG. 2D is a close up perspective view of the sled **200** when it is partially pulled away from the bay **128** and the lever mechanism **232** is in an open or released position.

[0037] FIGS. 2B and 2D are a perspective view of the example lever mechanism **232** on the sled **200** in respective closed and open positions. The lever mechanism **232** has a longitudinal body member **240** that includes a pivot end **242** and a release end **244**. The longitudinal body member **240** thus defines the lever of the lever mechanism **232**. The pivot end **242** is rotatably attached to a recessed support plate **246** on a side of the front end **210** of the sled **200**. The release end **244** is held via a latch mechanism **250** that extends from a support plate **252** that is part of the upper panel **212**. The release end **244** includes a perpendicular handle **254** to allow a user to better engage the lever mechanism **230** to rotate the longitudinal body member **240** around the pivot end **242**.

[0038] The latch mechanism **250** holds the lever mechanism **232** in the closed position in FIG. 2A via engaging a tab **256** that extends from the release end **244** as shown in FIG. 2D. When the latch mechanism **250** is released, the tab **256** is disengaged and thus the lever mechanism **232** is free to rotate away from the front end **210** of the sled **200** as shown in FIGS. 2C-2D. Spring force, as will be explained below, forces the lever mechanism **232** away from the upper panel **212**. The longitudinal body member **240** of the lever mechanism **232** may then be rotated by a user to apply force to assist in disconnecting the connectors and thus allow the sled **200** to be pulled out of the bay **128**.

[0039] FIG. 3 is a bottom view of the example lever mechanism **232** in FIGS. 2A-2D. The components of the lever mechanism **232** is fabricated from high strength metal or alloy and are preferably press stamped. The longitudinal body member **240** is construed of generally rectangular plates that form a lever. The pivot end **242** includes a support strut **310** that is attached to a side of a semi-circular spur gear **312**. The opposite side of the semi-circular spur gear **312** includes a semi-circular edge **314** with gear teeth. The semi-circular spur gear **312** includes a support cylinder **316** that has a hole **318** therethrough. The support cylinder **316** is joined to one end of the support strut **310**. A pin inserted through the hole **318** allows the semi-circular spur gear **312** to rotate on the support cylinder **316** and thus translate rotational motion from the longitudinal body member **240** of the lever mechanism **232**.

[0040] The semi-circular spur gear **312** is positioned within an ejector component **320**. FIG. 4 is a close up view of the semi-circular spur gear **312** in relation to the ejector component **320**. The ejector component **320** is a frame having a circular section **322** that has a hole **324** therethrough. A pin inserted through the hole **324** allows the ejector component **320** to be rotated. The ejector component **320** includes two angled support arms **330** and **332** that each have one end joined to the circular section **322**. The opposite end of the support arm **330** is a tab **334**. The opposite end of the support arm **332** is a tab **336**. The support arms **330** and **332** are joined together by a circular internal gear member **340** near the tabs **334** and **336**. The internal gear member **340** is semi-circular in shape and has an interior surface **342** with gear teeth. The gear teeth of the interior surface **342** engage the gear teeth of the semi-circular spur gear **312**. Thus, the circular section **322**, support arms **330** and **332**, and internal gear member **340** form the frame shape of the ejector component **320**. As shown in FIG. 4, the spur gear **312** is positioned within the frame shape formed by the circular section **322**, support arms **330** and **332**, and internal gear member **340**.

[0041] Returning to FIG. 3, the recessed support plate **246** is attached in perpendicular orientation to the upper panel **212**. The support plate **246** has a pin **350** for mounting and positioning the spur gear **312**. Thus, the pin **350** is inserted in the hole **318** of the support cylinder **316**. This allows the attached spur gear **312** to rotate around the pin **350**. A second pin **352** on the support plate **246** is positioned in proximity to the pin **350**. The circular section **322** of the ejector component **320** is attached to the support plate **246** via the pin **352**. The pin **352** is inserted through the hole **324** of the circular section **322** and thus the ejector component **320** rotates around the pin **352**. The positioning of the two pins **350** and **352** orients the spur gear **312** so the teeth of the semi-circular edge **314** mesh with the teeth on the interior surface **342** of the internal gear member **340** of the ejector component **320**. Thus, rotational motion from the spur gear **312** is translated into rotational motion of the ejector component **320**. Another pin **354** is positioned on the support plate **246** at a predetermined distance from the spur gear **312** and the ejector component **320**. A spring **356** has a first end **358** that is wrapped around the pin **354**. The coil of the spring **356** is wrapped around the pin **350**. The spring force of the spring **356** rotates the spur gear **312** in a counter-clockwise direction, and is translated to rotating the ejector component **320**.

[0042] The spring **356** is compressed when the lever mechanism **232** is in a closed position and held by the latch mechanism **250**. The ejector component **320** thus is fixed in place and cannot rotate counter-clockwise. The latch mechanism **250** is attached to a mounting bracket **370** that is attached to the interior side of the support plate **252**. The latch mechanism **250** extends through an aperture in the support plate **252** to hold the lever mechanism **232** in a closed position as shown in FIG. 2A.

[0043] FIG. 5 shows a perspective view of the lever mechanism **232** in relation to the respective latch mechanism **250** and the mounting bracket **370** that is joined to the upper panel **212** of the sled **200**. The mounting bracket **370** has a plate **510** and perpendicular sides **512** and **514** with respective tabs **516** and **518** that allow the mounting bracket **370** to be attached to the support plate **252**. The latch mechanism **250** is mounted on the mounting bracket **370** and oriented to extend from the support plate **252**. The latch mechanism **250** rotates to engage the tab **256** of the lever mechanism **232**. The latch mechanism **250** is secured to the mounting bracket **370** via a screw **520** that allows the latch mechanism **250** to be rotated. The latch mechanism **250** for the lever mechanism **230** in FIGS. 2A-3B is identical to the latch mechanism **250** shown in FIG. 5.

[0044] FIG. 6A-6C shows a sequence of the operation of the lever mechanism **232** in FIG. 2A that assists in releasing the sled **200**. FIG. 6A shows the lever mechanism **232** in the closed position where the sled **200** is entirely inserted into the chassis **110**. The tab **256** on the release end **244** is held by the latch mechanism **250**. Thus, the longitudinal body member **240** prevents the rotation of the ejector component **320** by locking the spur gear **312** in place. In this position, the spur gear **312** rests against the support arm **332** of the ejector component **320**. The tab **336** thus extends through the cutout **134** of the edge member **130**. The support arm **332** and corresponding tab **334** are angled

to position the tab **334** away from the flange **132** of the edge member **130**.

[0045] FIG. **6B** shows the lever mechanism **232** released by a user by rotating the latch mechanism **250**, thus releasing the lever mechanism **232**. The spring **356** causes the spur gear **312** to rotate counter-clockwise. The rotational movement of the spur gear **312** via meshing with the gear teeth on the interior surface **342** of the ejector component **320** causes the ejector component **320** to rotate counter-clockwise. The rotation of the spur gear **312** translates to the support strut **310** that causes the longitudinal body member **240** of the lever mechanism **232** to rotate counter-clockwise. Once the lever mechanism **232** is released, a user may pull the lever body away from the upper panel **212** via the handle **254**. The rotation of the longitudinal body member **240** causes the spur gear **312** to rotate and thus rotate the ejector component **320**. The rotation of the ejector component **320** causes the support arm **330** to rotate the tab **334** to contact the flange **132** of the side wall **120**.

Correspondingly, the support arm **332** rotates the tab **336** away from the cutout **134**.

[0046] After the lever mechanism **232** is released, a user may fully rotate the lever mechanism **232** such that the longitudinal body member **240** is approximately perpendicular to the front end **210** of the sled **200**. The position of the lever mechanism **232** after being fully rotated is shown in FIG. **6C**. The motion of the lever mechanism **232** causes the spur gear **312** to rotate counter-clockwise. As the spur gear **312** is rotating, the gear teeth on the semi-circular edge **314** mesh with the gear teeth on the interior surface **342** of the internal gear member **340**. As the spur gear **312** rotates, it contacts the support arm **330** of the ejector component **320**. The further rotation of the spur gear **312** translates motion via the meshing gear teeth and causes the ejector component **320** to rotate counter-clockwise. The rotation of the ejector component **320** causes the tab **334** to contact the flange **132**. The force applied on the flange **132** via the tab **334** causes the sled **200** to be pushed away from the bay. The force assists in unplugging the connectors at the rear of the sled **200** from the bay of the chassis. Once the connectors are disconnected, a user may slide the sled **200** all the way out of the bay **128**.

[0047] FIGS. **6D-6F** shows a sequence of the operation of the lever mechanism **232** to assist in inserting the sled **200** in the bay **128** of the chassis **110**. As shown in FIG. **6D**, the sled **200** is initially partially inserted into the bay **128**. The lever mechanism **232** is pulled out to the fully open position as shown in FIG. **6D**. The rotational position of the lever mechanism **232** causes the tab **334** to be rotated to a perpendicular position relative to the flange **132**. The sled **200** may be pushed into the bay to a position where the connectors of the sled are partially inserted in the corresponding chassis connectors. When the sled **200** is in such a position, the tab **334** contacts the flange **132**. Due to the rotational position of the ejector component **320**, the tab **336** is rotated to a position that does not interfere with the edge member **130** and allows movement of the sled **200** into the bay **128**.

[0048] Once the sled **200** is inserted in the bay **128** in the position shown in FIG. **6D**, the user will push the lever mechanism **232** clockwise toward the upper panel **212** as shown in FIG. **6E**. Rotating the lever mechanism **232** toward the upper panel **212** causes the spur gear **312** to rotate clockwise from the initial position in contact with the support arm **330**. This causes the ejector component **320** to rotate clockwise thus moving the tab **334** away from the flange **132** and moving the tab **336** into the cutout **134**. As the lever mechanism **232** is moved to the closed position, the tab **336** contacts the side of the cutout **134** and thus applies force against the side of the cutout **134** to push the sled **200** into the bay. The additional force applied through the lever mechanism **232** assists in fully connecting the connectors on the end of the sled **200** with the corresponding connectors in the chassis.

[0049] FIG. **6F** shows the sled **200** fully inserted into the bay **128** of the chassis. In this position, the lever mechanism **232** is locked in the closed position by the tab **256** being held by the latch mechanism **250**. The spur gear **312** has been fully rotated to rest against the support arm **332**. The additional travel of the spur gear **312** causes the ejector component **320** to rotate and thus translate force to linearly push the sled **200** into the bay **128**.

[0050] FIG. 7A is a perspective view of the latch mechanism **250** attached to the mounting bracket **370**. FIG. 7B is an exploded perspective view of the components of the latch mechanism **250** and the mounting bracket **370**. The latch mechanism **250** has a generally rectangular main body **710**. In this example, the main body **710** may be plastic, but other suitable materials may be used. The main body **710** includes a ridged edge **712** that allows a user to better contact the main body **710** to rotate the latch mechanism **250**. The main body **710** includes a contact edge **716** that includes a slot **718**. The slot **718** engages the tab **256** of the lever mechanisms such as the lever mechanism **232** in FIG. 2B. The main body **710** includes a circular hole **720** that is placed over a cylindrical support **730** that extends from the plate **510**. The cylindrical support **730** may be attached to the screw **520** that holds the main body **710** on the cylindrical support **730**. The main body **710** thus may rotate on the cylindrical support **730**.

[0051] The plate **510** has a hook **732** that allows attachment of one end of a coil spring **734**. The coil of the coil spring **734** is wrapped around the cylindrical support **730**. Thus, the spring force of the coil spring **734** rotates the main body **710** in a clockwise direction to lock the tab **256** of the lever mechanism. A user may rotate the main body **710** counter-clockwise to release the tab **256** and thus allow the lever mechanism to freely rotate.

[0052] FIG. 8A shows the latch mechanism **250** in the locked position that holds the lever mechanism **232** in the closed position. As shown in FIG. 8A, the spring force from the coil spring **734** forces the main body **710** to be rotated clockwise around the cylindrical support **730** and screw **520**. In this position, the slot **718** engages the tab **256** of the lever mechanism **232** and holds the lever mechanism **232** in place.

[0053] FIG. 8B shows the example latch mechanism **250** in the released position. A user may release the lever mechanism by pressing the ridged edge **712** and rotating the main body in a counter-clockwise direction. This compresses the coil spring **734**, but releases the tab **256** from the slot **718**, allowing the lever mechanism **230** to be rotated away from the sled **200**.

[0054] FIGS. 9A-9C show the sequence of removing the sled **200** from the server chassis **110** using the example lever mechanisms **230** and **232**. As shown in FIG. 9A, the lever mechanisms **230** and **232** have been released from the latch mechanisms **250**. The lever mechanisms **230** and **232** thus may be rotated to the open position to assist in disconnecting the sled **200** from the connectors in the bay **128** of the chassis **110**.

[0055] As the example lever mechanism **230** and **232** allow additional force to be applied to disconnect the connectors, the sled **200** may be pulled out of the bay **128** of the chassis **110** as shown in FIG. 9B. The lever mechanism **230** and **232** are still rotated outward.

[0056] Once the sled **200** is pulled out of the bay **128**, the lever mechanisms **230** and **232** are no longer needed and thus can be rotated to the closed position. The user thus locks the lever mechanisms **230** and **232** in the closed position by locking them with the latch mechanisms **250** as shown in FIG. 10C. The sled **200** may thus be transported without damaging the lever mechanisms **230** and **232**.

[0057] FIGS. 10A-10C show the sequence of inserting the sled **200** into the server chassis **110** using the example lever mechanisms **230** and **232**. As shown in FIG. 10A, the sled **200** is initially partially inserted in the bay **128** of the chassis **110**. The lever mechanisms **230** and **232** may be released so the lever mechanisms **230** and **232** rotate away from the sled **200**.

[0058] When the sled **200** is mostly inserted in the bay **128**, the lever mechanisms **230** and **232** may be rotated toward the closed position as shown in FIG. 10B. The lever mechanisms **230** and **232** allow additional force to be applied to push the connectors of the sled **200** to fully engage the connectors of the chassis **110** in the rear end of the bay **128**. Once the sled **200** is fully inserted in the bay **128**, the lever mechanisms **230** and **232** are rotated to the fully closed position and are locked in place by the respective latch mechanisms **250** as shown in FIG. 10C.

[0059] As used in this application, the terms “component,” “module,” “system,” or the like, generally refer to a computer-related entity, either hardware (e.g., a circuit), a combination of

hardware and software, software, or an entity related to an operational machine with one or more specific functionalities. For example, a component may be, but is not limited to being, a process running on a processor (e.g., digital signal processor), a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a controller, as well as the controller, can be a component. One or more components may reside within a process and/or thread of execution, and a component may be localized on one computer and/or distributed between two or more computers. Further, a “device” can come in the form of specially designed hardware; generalized hardware made specialized by the execution of software thereon that enables the hardware to perform specific function; software stored on a computer-readable medium; or a combination thereof.

[0060] The terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting of the invention. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Furthermore, to the extent that the terms “including,” “includes,” “having,” “has,” “with,” or variants thereof, are used in either the detailed description and/or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising.”

[0061] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art. Furthermore, terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0062] While various embodiments of the present invention have been described above, it should be understood that they have been presented by way of example only, and not limitation. Although the invention has been illustrated and described with respect to one or more implementations, equivalent alterations and modifications will occur or be known to others skilled in the art upon the reading and understanding of this specification and the annexed drawings. In addition, while a particular feature of the invention may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application. Thus, the breadth and scope of the present invention should not be limited by any of the above described embodiments. Rather, the scope of the invention should be defined in accordance with the following claims and their equivalents.

Claims

1. A lever mechanism for insertion to and removal of a sled from a chassis, the lever mechanism comprising: a longitudinal lever member having a release end and an opposite pivot end; a spur gear having gear teeth coupled to the pivot end of the longitudinal lever member, wherein the spur gear is rotated by rotating the longitudinal lever member; and an ejector component rotatably connectable to the sled, the ejector component having a frame defined by an internal gear support, a first support arm, and a second support arm, wherein the spur gear is positioned within the frame and engages the internal gear support to rotate the ejector component, wherein the first support arm includes a tab engaging a flange of the chassis to apply force to remove the sled from the chassis when the longitudinal lever member is rotated in a first direction and the second support arm includes a second tab engaging a cutout of the chassis to apply force to insert the sled in the chassis when the longitudinal lever member is rotated in a second direction.

2. The lever mechanism of claim 1, further comprising: a support plate rotatably attached to the spur gear and ejector component; and a spring having one end connected to the support plate and an opposite end connected to the spur gear, the spring biasing the spur gear, in the first rotational direction.

3. The lever mechanism of claim 1, wherein the release end of the longitudinal lever member includes a tab that engages a latch mechanism on the sled to hold the longitudinal lever member in place.
4. The lever mechanism of claim 3, wherein the latch mechanism rotates between a position engaging the tab and a position releasing the tab.
5. The lever mechanism of claim 1, wherein the release end of the longitudinal lever member includes a handle.
6. The lever mechanism of claim 1, wherein the spur gear is connected to a cylindrical support, and wherein the longitudinal lever member is connected to the cylindrical support.
7. The lever mechanism of claim 1, wherein the longitudinal lever member, the spur gear, and the ejection component generate force to disconnect connectors of the sled from connectors of the chassis when rotated in the first direction.
8. A sled for insertion into a bay of a chassis defined by two edge members each having a flange and a cutout, the sled comprising: a tray holding electronic components; a front panel coupled perpendicularly to the tray; and a first lever mechanism mounted on a side of the front panel, the first lever mechanism including: a longitudinal lever member having a release end and an opposite pivot end; a spur gear having gear teeth coupled to the pivot end of the longitudinal lever member, wherein the spur gear is rotated by rotating the longitudinal lever member; and an ejector component rotatably connected to the sled, the ejector component having a frame defined by an internal gear support, a first support arm, and a second support arm, wherein the spur gear is positioned within the frame and engages the internal gear support to rotate the ejector component, wherein the first support arm includes a tab engaging one of the flanges to apply force to remove the sled from the bay when the longitudinal lever member is rotated in a first direction and the second support arm includes a second tab engaging one of the cutouts to apply force to insert the sled in the bay when the longitudinal lever member is rotated in a second direction.
9. The sled of claim 8, further comprising a second lever mechanism mounted on an opposite side of the front panel, the second lever mechanism including: a longitudinal lever member having a release end and an opposite pivot end; a spur gear having gear teeth coupled to the pivot end of the longitudinal lever member, wherein the spur gear is rotated by rotating the longitudinal lever member; and an ejector component rotatably connected to the sled, the ejector component having a frame defined by an internal gear support, a first support arm, and a second support arm, wherein the spur gear is positioned within the frame and engages the internal gear support to rotate the ejector component, wherein the first support arm includes a tab engaging the other one of the flanges to apply force to remove the sled from the bay when the longitudinal lever member is rotated in a first direction and the second support arm includes a second tab engaging the other one of the cutouts to apply force to insert the sled in the bay when the longitudinal lever member is rotated in a second direction.
10. The sled of claim 9, wherein the first and second lever mechanisms generate sufficient force to disconnect connectors of the sled from connectors of the bay when rotated in the first direction.
11. The sled of claim 8, further comprising: a support plate rotatably attached to the spur gear and ejector component; and a spring having one end connected to the support plate and an opposite end connected to the spur gear, the spring biasing the spur gear in the first rotational direction.
12. The sled of claim 8, further comprising a latch mechanism on the front panel, wherein the release end of the longitudinal lever member includes a tab that engages the latch mechanism to hold the longitudinal lever member in place.
13. The sled of claim 12, wherein the latch mechanism rotates between a position engaging the tab and a position releasing the tab.
14. The sled of claim 8, wherein the release end of the longitudinal lever member includes a handle.
15. The sled of claim 8, wherein the spur gear is connected to a cylindrical support, and wherein

the longitudinal lever member is connected to the cylindrical support.

16. A computer system comprising: a chassis having two side walls, a shelf between the two side walls, and a bay defined by the two side walls and the shelf, each of the two side walls having edge members with longitudinal flanges; and a sled insertable in the bay of the chassis between the two side walls, wherein the sled includes a front panel, a tray, and a rotating lever mechanism on the front panel, the rotating lever mechanism including: a longitudinal lever member having a release end and an opposite pivot end; a spur gear having gear teeth coupled to the pivot end of the longitudinal lever member, wherein the spur gear is rotated by rotating the longitudinal lever member; and an ejector component rotatably connected to the sled, the ejector component having a frame defined by an internal gear support, a first support arm, and a second support arm, wherein the spur gear is positioned within the frame and engages the internal gear support to rotate the ejector component, wherein the first support arm includes a tab engaging one of the longitudinal flanges to apply force to remove the sled from the bay when the longitudinal lever member is rotated in a first direction and the second support arm includes a second tab engaging a cutout on one of side walls to apply force to insert the sled in the bay when the longitudinal lever member is rotated in a second direction.
